

Ask Before You Transmit: An Energy-Frugal Evidence-Level Semantic Communication System for UAV Visual Question Answering

Qiankun Zhang, Yue Liu, *Member, IEEE*, and Zhijin Qin, *Senior Member, IEEE*

Abstract

Answering natural-language questions about a live aerial feed is an increasingly common unmanned-aerial-vehicle (UAV) service, served today by an energy-oblivious default: upload the image, run a vision–language model (VLM). We measure what that default costs per *answered question*—transmit plus inference energy—and show that most of it is avoidable. On a 548-image aerial pool whose 936 held-out test questions are replayed over five question types, six SNRs, and three fading channels (16,848 scored decisions), plus a 490-image second pool replayed Rician-only (710 questions, 4,260 decisions)—together, to our knowledge, the largest end-to-end VQA-accuracy evaluation of a real VLM receiver over coded wireless links—a task-agnostic transmitter feeding a frozen commercial VLM exhibits a stable *evidence–question complementarity*: symbolic reasoning questions (counting, comparison, co-presence) are answered *more* accurately from ~ 0.9 KB detector tokens than from the transmitted image (+0.11 to +0.17 accuracy), perception questions need the image (−0.08), and the fifth type (threshold) is a boundary type whose sign flips across datasets. A zero-parameter rule (symbolic→token, perception→image) is statistically indistinguishable from a data-calibrated selector, and the token branch must be decoded *symbolically*: handing the same tokens to the VLM as text costs ≈ 15 points. Because the level decision also decides *whether the VLM forward pass runs at all*, routing is an energy decision: under a joint communication-plus-computation accounting with GPU power measured on the campaign workload, the per-answer energy frontier is *compute-dominated*, and type-aware routing cuts measured joules per answered question by $2.2\times$ against rate-adaptive image transmission at every SNR while *raising* accuracy (0.67 vs. 0.60 on every channel; ≈ 0.68 vs. ≈ 0.61 against even error-free image transfer); fixed tokens are $75\text{--}79\times$ cheaper than the image ($33\text{--}35\times$ cheaper than routing), and an energy price swept over a per-sample correctness predictor makes the router *energy-controllable* between

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Q. Zhang and Y. Liu are with the Faculty of Applied Sciences, Macao Polytechnic University, Macao SAR, China (e-mail: yue.liu@mpu.edu.mo).

Z. Qin is with the Department of Electronic Engineering, Tsinghua University, Beijing, China.

the all-token and accuracy-first endpoints. A conditional energy–accuracy Pareto-dominance proposition formalizes the effect, a non-monotone token-budget effect (+7.7 points from truncation) sharpens it, and the rule needs no channel-state input. In task-oriented aerial VQA, energy efficiency comes from asking, *before* transmitting, which evidence the question actually needs.

Index Terms

Green communications, energy efficiency, semantic communication, visual question answering, UAV networks, task-oriented communication, evidence routing, vision–language models.

I. INTRODUCTION

UNMANNED aerial vehicles (UAVs) have become a standard sensing platform for low-altitude inspection, traffic monitoring, and disaster response. In an increasingly common interaction pattern, a UAV streams a live aerial feed while a ground operator, or an automated decision loop, poses natural-language *questions* about the scene—“how many vehicles are on the bridge?”. Answering them is a visual question answering (VQA) task, and modern vision–language models (VLMs) answer well when the image reaches the receiver. The obstacle is no longer the answerer but the cost of feeding it: the UAV is battery-limited, the air-to-ground channel is bandwidth-limited and error-prone, and the edge VLM forward pass is the most energy-hungry single step in the loop—in our measurements it costs more than transmitting the image at every link we test (Section VI-F). A full image per query thus spends the system’s two scarcest resources—transmit and edge-inference energy—on questions that mostly do not need the image at all.

Semantic and task-oriented communication offers a way out: transmit only what the downstream task needs [1]–[3]. A green branch of this literature makes energy the explicit objective, up to joint communication-plus-computation budgets [4]–[6]; for VQA specifically, task-specific transceivers are trained jointly so answer semantics survive the channel [7], [8]. Yet two assumptions persist: the image (or its learned latent) is treated as the *accuracy upper bound*, cheaper representations being only a latency-for-accuracy trade [8]; and the transmitter and receiver are co-designed and co-trained, coupling the result to one reasoning backend and precluding a frozen commercial VLM.

This paper starts from a different question, asked *at transmission time*: given *this* question, which *level* of evidence does the answer actually require? We instantiate a small ladder of evidence levels—a cached answer (s_0), symbolic detector tokens (s_1), a rate-adaptive coded image (s_2), and an optional region-of-

interest crop (s_3)—produced by a *task-agnostic* transmitter (an off-the-shelf detector plus standard channel coding) and consumed by a *frozen* commercial VLM receiver. Because the level decision selects not only the payload but *which computation runs*—a few-operation CPU symbolic decode versus a VLM forward pass—evidence routing is simultaneously an energy decision; we account every answered question in joules, transmit plus compute (Section III-G). The head-to-head evaluation concerns the two transmit-bearing levels (s_1, s_2); s_0 and s_3 are design/oracle-headroom options (Sections IV-D and VII-A). Routing among these levels by question type turns out to be surprisingly effective and analytically well-founded—exact in an idealized limit, directional outside it.

A. Contributions

We make three contributions.

(1) An evidence–question complementarity and a zero-parameter routing rule. On two aerial benchmarks we find a stable pattern: symbolic-reasoning questions (counting, comparison, co-presence) are answered *more* accurately from compact detector tokens than from the transmitted image (+0.11 to +0.17 accuracy), whereas perception questions (presence) need the image (−0.08). A zero-parameter rule (symbolic→token, perception→image) is *statistically indistinguishable* from a data-calibrated Wilson lower-confidence-bound (LCB) selector: 0.6698 vs. 0.6686 pooled over the three channels (paired McNemar $p=0.12$; bit-for-bit identical on AWGN, one deviating low-SNR presence cell per fading channel) and within noise on a second dataset ($p=0.20$; Section IV-B). Question type is the *only* load-bearing routing dimension—SNR, viewpoint, freshness, and risk contribute at most ± 0.003 pooled accuracy to the level decision (Section VI-L)—and the token branch must be decoded *symbolically*: a VLM handed the same tokens as text loses ≈ 15 points (Section VI-K).

(2) A large-scale, unified evaluation methodology—channel uses *and* joules—for real VLM receivers over coded wireless links. We evaluate a 548-image primary pool—936 held-out test questions replayed over five question types, six SNRs, and three channels (AWGN, Rayleigh, Rician): 16,848 scored decisions—plus a 490-image second pool (710 test questions, 4,260 Rician-only decisions; Section VI-C)—with a *unified complex-channel-use* accounting comparing a baseline bank—cached-answer reuse, fixed-rate and rate-adaptive digital, analog and compact learned joint source–channel coding, error-free transfer—on one axis. On top we add a measured *joint communication-plus-computation energy* axis (Section III-G): GPU power measured on the campaign workload itself, every method charged joules per answered question, yielding an energy–accuracy frontier for the whole bank (Section VI-F). To our knowledge

this is the largest-scale end-to-end VQA evaluation of a real VLM receiver over coded links—not the first, credited to [9]; our increment lies on the scale, aerial-domain, evidence-routing, and measured-energy axes (Section II).

(3) Three counter-intuitive findings and a consistency check. (i) Adaptive evidence routing beats full-image transmission *even over an error-free channel* (≈ 0.68 vs. ≈ 0.61): the gain comes from evidence selection itself, not channel robustness. (ii) The measured per-answer energy frontier is *compute-dominated*: at every link we test, the VLM forward pass costs more than the radio, so the biggest energy lever available to the transmitter is not power or rate adaptation but *declining the VLM inference when cheap evidence suffices*—type-aware routing cuts the measured joint energy per answered question by $2.2\times$ relative to rate-adaptive image transmission at every SNR while being *more* accurate, and the fixed token policy is $75\text{--}79\times$ cheaper than the image ($33\text{--}35\times$ cheaper than routing), a frontier an energy-price sweep makes controllable (Section VI-G). (iii) Confidence-ordered top- t token truncation *raises* accuracy on threshold questions by 7.7 points, acting as a false-positive filter—a non-monotone “less is more” effect absent from reconstruction-oriented progressive schemes. Because question type, not channel state, carries the decision, a policy-assumed-SNR $\times$ true-SNR mismatch matrix is essentially flat—no cliff from scheduling-layer CSI error; a sender-side per-sample predictor adds 0.7–1.1 points while nearly halving payload.

II. RELATED WORK

A. Green and energy-aware semantic communication

A dedicated line of work in this journal minimizes semantic-communication energy. Zheng *et al.* [4] allocate resources for energy-efficient semantic transmission in aerial-aided edge networks; Dai *et al.* [5] minimize the *joint* communication-plus-computation energy of a multi-modal probabilistic semantic system under latency and compression-ratio constraints—the joint energy accounting we also adopt (Section III-G); and Xu *et al.* [6] minimize system energy for rate-splitting multiple access with probabilistic knowledge graphs. These works *optimize* energy for a fixed semantic representation whose task quality enters as an analytical surrogate. We are complementary on two axes: we *measure* the task outcome—per-question VQA accuracy of a real frozen VLM over coded links, priced as a measured per-answer joint-energy frontier (Section VI-F)—and our lever is *which evidence level to transmit at all*, hence whether the dominant energy term, VLM inference, runs at all (Table III).

B. Goal-oriented VQA semantic communication

MU-DeepSC [7], [10] pioneered VQA as the receiver task with a jointly trained transceiver, its gains concentrated in low-SNR robustness over separate coding. GO-SG [8] ranks bounding boxes and scene-graph triplets by question keywords and observes—consistent with our finding—that evidence formats and question categories are complementary; however it treats full-image transmission as the accuracy *upper bound* (trading 4% accuracy for 65% latency), offers manual guidelines rather than an automatic selector, and answers with a neuro-symbolic reasoner, not a real VLM. We show instead that question-conditioned routing *surpasses* full-image transmission even error-free, via an automatic zero-parameter rule. Closest in task is VideoQA-SC [11], which learns an end-to-end bandwidth-adaptive spatiotemporal codec answering video questions without reconstruction. We differ in kind: a *task-agnostic* off-the-shelf detector plus a *standard* coding stack (LDPC, BPSK, rate-adaptive JPEG) feeding a *frozen commercial* VLM, contributing which evidence *level* to route per question rather than a learned codec—which could join the ladder as an additional learned compact-evidence level, a bandwidth-adaptive alternative to the coded image s_2 , within the same framework (cf. Limitations).

C. Task-adaptive modality selection

Park and Yoon [12] select visual semantic modalities per classic computer-vision task with a fixed task-to-semantics mapping and a diffusion-based receiver; VQA is not among the tasks and no per-question adaptation is learned. Large-multimodal-model-based vehicle networks [9] evaluate a real LLaVA receiver over fading channels but restrict question awareness to region selection *within* the image modality on 41 road-scene images. We instead route among evidence *levels* per question type on an order-of-magnitude larger aerial benchmark with unified channel-use accounting; we claim—carefully—the largest-scale and first evidence-level-routing evaluation, not the first real-VLM-over-wireless one.

On the token-budget axis, progressive edge–cloud VLM communication [13] encodes nested-prefix latent tokens with provably monotone reconstruction risk, and the scene-graph filtering of [12] reports that a pruned graph can slightly beat the full graph. Our top- t budget is confidence-ordered *symbolic* evidence: its sensitivity separates question families by an order of magnitude ($t=3$ vs. $t=48$), and moderate truncation *improves* accuracy (+7.7 on threshold) as a false-positive filter.

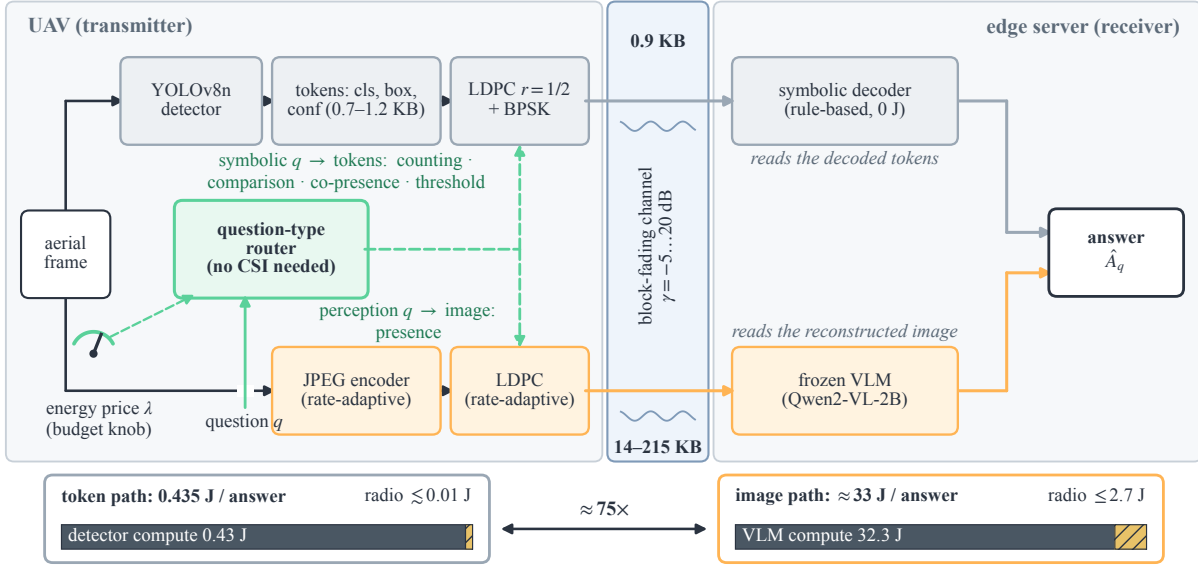


Fig. 1. System overview. The question-type router selects the evidence level *before* anything is transmitted: symbolic questions send the detector’s token packet (0.7–1.2 KB; LDPC $r=1/2$, BPSK), the perception question sends a rate-adaptive JPEG+LDPC image (14–215 KB delivered); s_0/s_3 are conditional options (Section III-A). Since tokens are answered by a rule-based symbolic decoder and images by the frozen VLM, routing also selects *which computation runs*—measured per-answer energy (bottom chips; $P_{tx}=0.5$ W, incremental compute): 0.435 J (token, detector-dominated) vs. ≈ 33 J (image, VLM-dominated), a $\approx 75\times$ gap. No CSI is needed at decision time; the energy price λ (Section VI-G) makes the operating point budget-tunable.

D. Adaptive coding and semantic-aware resource control

Predictive and adaptive deep coding [14] trains an OracleNet to predict reconstruction PSNR and select a transmission rate in the *reconstruction* domain; deep joint source–channel coding [15] and attention-modulated variants [16] likewise optimize fidelity, always with true SNR fed to the encoder. Our per-sample predictor (Section IV-C) operates in the *task-accuracy* domain from sender-only features and selects the evidence *level*; our CSI-mismatch analysis (Section VI-L) measures an axis these works never expose. On the control side, context-aware UAV semantic communication [17] and multidomain adaptive semantic communication [18] adapt compression or rate inside a control loop; neither performs question-conditioned evidence selection, and the resource-control coupling is pursued in a companion study.

III. SYSTEM MODEL

We consider a UAV that observes a low-altitude scene and an edge server that answers ground-user VQA queries about the live feed (Figure 1). The transmitter does *not* aim to reconstruct the image: an on-board object detector (YOLOv8n [19], fine-tuned on the target aerial domain) and an evidence selector map the current question to a level $s \in \{s_0, s_1, s_2, s_3\}$, and only that evidence crosses the physical layer and wireless channel to the edge receiver, which decodes symbolic evidence or feeds a reconstructed image to a frozen VLM (Qwen2-VL-2B [20]) to produce the answer.

A. Evidence levels

The ladder comprises four levels. Level s_0 (cache) returns a stored answer under a freshness constraint and transmits nothing (0 B). Level s_1 (token) transmits the detector output—class labels, boxes, and confidences—as a per-frame symbolic packet (0.7–1.2 KB over LDPC $r=1/2$, BPSK). Level s_2 (image) transmits a rate-adaptive JPEG bitstream (~ 110 KB typical). Level s_3 (ROI) transmits a detection-guided crop (coded as s_2), supported by the framework but outside the main comparison.

B. Symbolic-token physical layer (s_1)

Detector tokens are packetized per frame and protected by a rate-1/2 LDPC code [21] with BPSK modulation (0.5 bit/use), so the accounting charges 16 complex uses per byte. Two channel-loss models are used. In the *rate-adaptive* mode—the one behind every headline result—the per-frame loss probability is the *information-outage* probability $\Pr[\log_2(1 + \gamma|h|^2) < r_{\min}]$ estimated by Monte-Carlo over the fading distribution (not an optimistic closed-form ergodic value); a separate fixed-rate “naive” baseline draws its frame-error rate (FER) from an *actual* LDPC/BPSK encode–decode Monte-Carlo over block-fading realizations. On a frame loss, with probability 0.5 the frame is dropped, otherwise the payload is garbled (labels to unknown, boxes jittered); a corrupted token still carries partial counting evidence, so s_1 degrades gracefully.

On finite blocklength and honesty of the token cost. The token frame is short ($n \approx 96$ –150 complex uses), so billing it at the nominal $r=1/2$ rate ignores a ≈ 0.8 –1 dB dispersion penalty [22]: token airtime is mildly *under-billed* at low SNR, inflating the already enormous $265\times$ image-to-token gap by at most $\approx 1.3\times$; the s_1 *accuracy* is read from the Monte-Carlo outage model and carries no capacity-formula optimism.

C. How each evidence level is answered

A point that governs every result: the two evidence levels are answered by *two different decoders*, the VLM serving the image path only. Level s_2 (image): the frozen VLM is prompted with the received (degraded) image and the question, decoded *greedily* (temperature 0, at most 24 new tokens). Level s_1 (token) is *not* sent to the VLM: a lightweight *symbolic decoder* answers by a deterministic rule per family (*presence* = “yes” iff the received target count exceeds 0; *counting* = the calibrated count; *comparison/co_presence/threshold* = the corresponding predicate over per-class counts, “>”, “both > 0”,

“ $\geq N$ ”). No neural network runs on the token path, so the claim is best read as *evidence-pipeline* complementarity—each decoder the natural answerer for its format.

Count calibration. For counting, the raw detector count is scaled by a single per-(class,channel) multiplicative factor—the clipped ratio of ground-truth to detected counts—applied only when the raw count is at least 3. We report honestly that this scalar is estimated on the pooled evaluation set rather than a disjoint fit split; it is applied identically to the calibrated-selector and zero-parameter-rule pipelines, so it cannot manufacture a gap *between* them, and a disjoint-fit check bounds the optimism: refitting the scale on the 80% split alone moves hold-out counting accuracy from 0.451 to 0.448 and the three-channel counting gap of record (Fig. 7) from +0.171 to +0.168—no conclusion changes.

Judging. Yes/no answers map synonyms to a canonical {yes,no} and are graded by exact match; counting answers extract the leading integer, graded within $\pm 10\%$ (floor ± 1) of ground truth.

Detector training. The YOLOv8n detector is fine-tuned on the official VisDrone-DET *training* split (6,471 images); the 548 evaluation images are the disjoint official *validation* split, so the detector never sees any evaluation image (or its held-out 20%).

D. Rate-adaptive image physical layer (s_2)

For s_2 , JPEG quality is matched to the achievable spectral efficiency: the byte budget $B \tau \text{SE}(\gamma, \text{channel})/8$, with SE estimated as $\mathbb{E}[\log_2(1 + \gamma|h|^2)]$ over the fading distribution, is fitted by the highest JPEG quality in [8, 95] (reducing resolution only if $q=8$ overflows), the bits are LDPC-coded, and a deep fade that cannot meet a minimum payload yields a flat-gray frame. This is the classical separate source–channel coding (SSCC) baseline; transmitting at a *fixed* rate instead (the fixed-rate image baseline of Section VI) produces the digital cliff at low SNR.

E. Channels

We evaluate three channels [23]: AWGN, Rayleigh fading (a no-LoS worst-case bound), and Rician fading with $K=6$ dB—within the $K \approx 2\text{--}15$ dB range measured for UAV air-to-ground links [24]. The elevation-dependent LoS geometry follows Al-Hourani *et al.* [25], and SNR is swept over $\gamma \in \{-5, 0, 5, 10, 15, 20\}$ dB.

Block-fading and ergodicity. At 14 m/s ground speed and 2.4 GHz carrier, $f_D \approx 112$ Hz and $T_c \approx 3.8$ ms: an LDPC codeword of 96 complex uses spans $\approx 96 \mu\text{s} \ll T_c$ (block fading holds), while the 0.3 s

transmission window covers ≈ 79 coherence blocks, justifying the ergodic spectral-efficiency accounting used for s_2 .

F. Unified channel-use accounting

To compare analog and digital methods on a single axis, every method is billed in *complex channel uses* per query: for a payload of b bits, $u_{s_2} = b/\text{SE}(\gamma, \text{channel})$ and $u_{s_1} = b/0.5$, the fixed-rate digital baseline also using the 0.5 bit/use denominator. An analog (uncoded) baseline consumes its native uses; one time slot of bandwidth B and duration τ provides $B\tau = 1 \text{ MHz} \times 0.3 \text{ s} = 3 \times 10^5$ uses (this single-UAV study freezes the service profile; fleet-level bandwidth sharing is deferred to a companion scheduling study). As a concrete scale comparison at -5 dB Rayleigh: rate-adaptive full-image transmission costs 3.72×10^6 uses (12.4 slots), the fixed-rate image a constant 5.28×10^6 , an analog baseline a constant 3×10^5 , while symbolic tokens cost only $\sim 1.4 \times 10^4$ uses—a $265\times$ gap between image and token.

G. Joint energy accounting

Channel uses measure spectral cost, but the operational cost of a query on a battery-limited UAV is energy, and the chosen evidence level also determines *which computation runs*. We therefore adopt a joint communication-plus-computation accounting in the spirit of PSCoM [5], charging every answered question

$$E(s, \gamma) = \underbrace{\frac{u_s(\gamma)}{B} P_{\text{tx}}}_{E_{\text{tx}}} + \underbrace{E_{\text{cmp}}(s)}_{\text{answerer compute}} \quad [\text{J/answer}], \quad (1)$$

where $u_s(\gamma)$ is the complex-channel-use cost of Section III-F (so u_s/B is airtime at bandwidth $B=1 \text{ MHz}$) and P_{tx} is the UAV transmit power, parameterized over 0.1–1 W—bracketing the 23 dBm (0.2 W) aerial-UE class of 3GPP TR 36.777 [26]—with 0.5 W as the headline point and a sensitivity sweep in Section VI-F.

The compute term follows the answering mechanism of Section III-C. Level s_2 invokes the frozen VLM at the edge; we *measure* its energy per answer on an RTX 4060, declared as an edge-server proxy (measurement protocol and numbers in Section VI-F). Level s_1 invokes the onboard detector, whose energy we parameterize as a Jetson-class embedded-GPU band (module power 7–15 W [27] times a 15–50 ms/frame latency band), keeping our own desktop-GPU detector measurement as a cross-check; the symbolic decoder is a few arithmetic operations on a CPU and is billed 0 J (declared, not measured). Level s_0 transmits and computes nothing. One billing asymmetry is declared: the token path pays for its

onboard encoder (the detector, 0.43 J mid-band) while the image path’s onboard source encoding—JPEG, or the DJSCC neural encoder—is billed 0 J; this is conservative *against* the token path and mildly favors the DJSCC baseline.

Scope. We bill the *marginal* (incremental-over-idle) compute energy as the headline—the energy a query adds to an already-powered platform—and report the total-power variant alongside. The compute-dominance finding (Section VI-F) is scoped to *unbatched, single-tenant* edge inference—one live query, one VLM forward pass. Propulsion/hover energy (invariant across transmission schemes) and common receiver-side radio energy are excluded. The finite-blocklength under-billing of token airtime (Section III-B, $\approx 1.3\times$) carries over to E_{tx} and equally cannot affect any conclusion: the image-to-token gaps of Section VI-F exceed two orders of magnitude.

H. Tasks, data, and split

The primary benchmark uses 548 real aerial images from VisDrone-DET [28]; the second uses 490 images from DroneVehicle [29]. Five question types are generated deterministically: *presence*, *counting*, *comparison* (two counts), *co_presence* (joint existence), and *threshold* (count-vs-threshold). Test sets are a 20% hold-out by `crc32` hash of `image_id`—deterministic and leakage-free; all reported numbers are test-set.

Answer-prior balance. Yes/no generation is explicitly balanced so accuracy cannot be gamed by a constant answer. Counted over the full question pool of the 548-image primary benchmark (train and test, before the hold-out): comparison is exactly 269/269 yes/no, threshold 274/274, *co_presence* 269/279, and presence 1186/961; counting contributes 1114 questions (214 of them in the 20% hold-out). A trivial “always-yes” predictor thus scores 0.49–0.55 per yes/no type, well below every method in Section VI.

Side features. Beyond the question type, each task carries three context features used to probe (and, in Section VI-L, to rule out) their influence on routing: a *viewpoint-quality* bin (good/medium/poor), a *risk* level (normal/critical), and a *freshness* bin (fresh/stale/expired; affecting only the cached level s_0). The selectors consuming these features—a Wilson-LCB look-up table (LUT), a zero-parameter semantic rule, and a per-sample parametric predictor—are detailed in Section IV.

IV. QUESTION-CONDITIONED EVIDENCE ROUTING

The three selectors—a calibrated LCB look-up table (LUT), a zero-parameter rule, and a per-sample predictor—all operate at the transmitter and require no channel-state feedback for the level decision.

A. Wilson-LCB calibrated selector

For each cell defined by (question type, SNR bin, and, optionally, viewpoint/freshness/risk) and each level s , we estimate answer accuracy on the training split with k successes of n trials and take its Wilson lower confidence bound $\text{LCB}_\alpha(k, n) = (\hat{p} + \frac{z^2}{2n} - z\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}) / (1 + \frac{z^2}{n})$, $\hat{p} = k/n$, $z = z_{1-\alpha}$ [30]. The selector picks $\arg \max_s \text{LCB}_\alpha$, guarding against sparse cells reading spuriously high; this is the *evidence routing* method of Section VI, *learned on the training split and reported on the test split*.

B. Zero-parameter semantic rule

The ablation of Fig. 9 (Section VI-L) shows that question type is the only routing dimension that carries weight—further conditioning changes pooled test accuracy by at most ± 0.003 —motivating a rule with *no free parameters*:

$$s^*(q) = \begin{cases} s_1 \text{ (token),} & q \in \{\text{counting, comparison,} \\ & \text{co_presence, threshold}\} \text{ (symbolic),} \\ s_2 \text{ (image),} & q = \text{presence (perception).} \end{cases} \quad (2)$$

Pooled over the three channels the rule (0.6698) and the calibrated LCB selector (0.6686) are *statistically indistinguishable* (paired McNemar $p=0.12$, 167 discordant of 16,848; fixed-token 0.6341, fixed-image 0.6040, oracle 0.7459, with CIs in Section VI-C and per-channel values in Table V, whose LUT column is calibrated per channel rather than on the pooled table). On AWGN the two are *bit-for-bit identical*—the same level in every calibration cell—while on each fading channel the learned policy deviates from the rule in exactly one cell (presence at -5 dB); on the second dataset the rule (0.7293) is nominally above the selector (0.7246) but not separably so (paired McNemar $p=0.20$, 224 discordant of 4,260). Section V explains the split: tokens preserve the sufficient statistic for discrete counting, the image preserves existence recall.

C. Per-sample parametric predictor

The rule is question-conditioned but sample-blind. We upgrade it with a logistic-regression predictor $P(\text{correct} \mid x, s)$ trained on *sender-available* features x (question type, target class, raw detection count, SNR, task metadata) using the OracleNet recipe [14] with a BCE (rather than MSE) objective. At inference we pick $\arg \max_s P(\text{correct} \mid x, s)$ per sample. This is evaluated as an offline re-selection over existing

logs and yields +0.7 to +1.1 points over the LUT across all three channels (recovering 10–15% of the oracle gap) while nearly *halving* payload (111 $\rightarrow$ 58 KB), by routing *presence* to tokens when sender-side detection evidence already suffices; gains concentrate on presence (+1.5 to +2.4 pt) and counting (+1.2 pt), exactly where per-sample information should help. The predictor must be sender-side for a measured reason: the receiver VLM’s own *verbalized* confidence is mildly *anti*-calibrated (AUC 0.406, below chance), so it would actively mislead the router.

D. ROI (s_3) as a conditional-routing option

The detection-guided ROI crop (s_3) is retained as a *conditional routing option* and an oracle-headroom probe rather than a headline mode: crops help perception questions that hinge on a small region yet hurt counting/comparison, which need the whole scene, so no static rule wins; s_3 stays out of the main comparison as an option the router *may* exercise when per-sample evidence localizes the answer.

Finally, for symbolic questions the token payload can be truncated to the top- t detections by confidence—a nested-prefix budget knob quantified in Section VI-N.

V. ANALYTICAL SUPPORT FOR EVIDENCE–QUESTION MATCHING

The zero-parameter rule sends symbolic questions as tokens and perception questions as images. This section gives two conditional results that make the split intelligible information-theoretically, stating their idealizing assumptions; the empirical gaps of Section VI carry the claim, and the analysis explains their *direction*.

A. Setup

Let I denote the source image and let the detector produce a token message $Z = g(I) = \{(c_j, b_j, p_j)\}_{j=1}^J$ of class labels, boxes, and confidences. Level s_1 transmits Z over the token physical layer (Section III-B); level s_2 transmits a lossy-coded image $\hat{I} = \kappa(I)$, an operation that factors through I . For a question q with ground-truth answer A_q , an *answerer* is a (possibly stochastic) map from the received evidence to $\hat{A}_q$.

B. Counting: a sufficiency lemma in the noiseless-detection limit

Call a detector a *noiseless counter* for target class c at operating point θ if the number of above-threshold class- c detections equals the true count, $\hat{N}_c(Z) = N_c(I)$ almost surely.

Lemma 1 (Counting sufficiency). *Consider a counting question with answer $A_q = N_c(I)$. For a noiseless counter, the transmitted count $\hat{N}_c(Z)$ is a sufficient statistic for A_q : the token path preserves A_q exactly, whereas the image path returns A_q only up to the information retained by the lossy map κ . Formally, since $A_q = \hat{N}_c(Z)$ is deterministic while $A_q \rightarrow I \rightarrow \hat{I}$ is a Markov chain, the data-processing inequality gives*

$$I(A_q; \hat{N}_c(Z)) = H(A_q) \geq I(A_q; \hat{I}), \quad (3)$$

with equality iff $\hat{I}$ also determines A_q . Hence no image-based answerer can strictly exceed the (exact) count-based answerer, and it is strictly worse whenever lossy coding makes $\hat{I} \not\rightarrow A_q$.

Proof. A noiseless counter attains $I(A_q; \hat{N}_c(Z)) = H(A_q)$, the maximum any statistic can, while DPI on $A_q \rightarrow I \rightarrow \hat{I}$ bounds $I(A_q; \hat{I}) \leq H(A_q)$, i.e. (3); equality requires $\hat{I}$ to determine the discrete count, which a rate-limited lossy code does not guarantee. $\square$

Remark 1 (Detection noise). *With a realistic detector, $\hat{N}_c(Z) = N_c(I) + \eta$, and Lemma 1 degrades to a comparison of noise sources: the token path loses only through the detector error η , the image path through both VLM recognition error and channel/compression distortion of $\hat{I}$. Tokens remain preferable whenever the detector’s counting noise is smaller than the VLM’s counting error on the degraded image—exactly the regime our measurements exhibit (counting token 0.45 vs. image 0.28); we treat this as directional, and it extends to comparison, co_presence, and threshold, whose answers are monotone in per-class counts.*

C. Perception: a false-alarm-constrained detection bound

Existence (presence) questions are recall-limited, and here the ordering reverses. We phrase it as a Neyman–Pearson-style bound so the comparison is fair (equal false-alarm budget on both paths).

Lemma 2 (Presence detection under a false-alarm constraint). *Fix a false-alarm budget α , i.e. $\Pr[\hat{A}_q = \text{yes} \mid N_c(I) = 0] \leq \alpha$, and let the token path answer “yes” iff at least one class- c token survives. Its detection probability is at most the detector’s operating-point recall, $P_D^{\text{tok}} \leq \rho(\theta)$: any instance the detector misses (or the channel drops) is unrecoverable from Z . An image answerer retains sub-threshold pixel evidence in $\hat{I}$ and can, at the same α , realize a $P_D^{\text{img}}(\alpha)$ not upper-bounded by $\rho(\theta)$; whenever $P_D^{\text{img}}(\alpha) > \rho(\theta)$, the image path attains strictly higher presence accuracy, and the gap widens as the token evidence is thinned (smaller top- t , coarser θ).*

Proof. A token “yes” requires a surviving above-threshold detection, so $P_D^{\text{tok}} \leq \rho(\theta)$ at false-alarm $\leq \alpha$ (channel loss only lowers it); $\hat{I}$ preserves sub-threshold evidence discarded by g , so the image-optimal test at the same α has $P_D^{\text{img}}(\alpha) \geq P_D^{\text{tok}}$, strictly whenever a missed instance leaves recoverable pixel evidence. $\square$

Remark 2 (Truncation is non-monotone). *Lemma 2 also predicts the token-budget effect: shrinking top- t lowers recall, hurting recall-limited presence, while raising precision on yes-biased questions—so truncation’s net accuracy effect is non-monotone and type-dependent, as observed (+7.7 points on threshold at $t=32$, Section VI-N).*

Corollary 1 (Direction of the rule). *In the idealized regimes of Lemmas 1 and 2, routing count-derived symbolic questions to s_1 and recall-limited perception questions to s_2 is the accuracy-maximizing level choice—exactly the zero-parameter rule of Section IV-B; outside the idealization it is a directional argument that the measurements of Section VI confirm.*

D. Energy–accuracy Pareto dominance of type-aware routing

The same structure yields a statement about the *joint energy* per answered question (Section III-G). Let T be the set of question types with prior mass w_t , let $a_s(t)$ denote the accuracy of level $s \in \{s_1, s_2\}$ on type t , and let $e_s(t)$ denote the per-answer energy of level s (transmit airtime plus the compute of the answerer that level invokes). A *type-measurable policy* is a map $\pi : T \rightarrow \{s_1, s_2\}$ with accuracy $A(\pi) = \sum_t w_t a_{\pi(t)}(t)$ and energy $E(\pi) = \sum_t w_t e_{\pi(t)}(t)$; the *fixed* policies π_{s_1}, π_{s_2} transmit one level for every question. We use two measured premises:

- (E1) *Pointwise energy ordering*: $e_{s_1}(t) < e_{s_2}(t)$ for every t —the token path is cheaper in airtime at every SNR (Section III-F) *and* in compute, since it declines the VLM forward pass, under the joint accounting of Section III-G (Table III; 0.43 vs. 32.3 J).
- (E2) *Complementarity with strict perception preference*: $a_{s_1}(t) \geq a_{s_2}(t)$ for symbolic $t \in T_{\text{sym}}$ and $a_{s_2}(t) > a_{s_1}(t)$ for perception $t \in T_{\text{perc}}$ (the sign pattern of Fig. 7).

Proposition 1 (Energy–accuracy Pareto dominance). *Under (E1)–(E2), the zero-parameter rule $r (T_{\text{sym}} \rightarrow s_1, T_{\text{perc}} \rightarrow s_2)$ satisfies, over the class of type-measurable policies: (i) $A(r) = \max_{\pi} A(\pi)$; (ii) among accuracy-maximal policies, r has the minimum energy E ; and (iii) for every accuracy target $\tau \in (A(\pi_{s_1}), A(r)]$, the rule is feasible for the constraint $A(\pi) \geq \tau$ and attains a per-answer energy no*

TABLE I
COMPARISON METHODS; THE SAME NAMES APPEAR IN ALL FIGURES AND TABLES.

Method	Meaning / paradigm
Error-free image	ideal error-free image transfer (reference, <i>not</i> an upper bound)
Fixed-rate image	fixed-rate LDPC digital image (digital cliff)
Rate-adaptive image	rate-adaptive JPEG+LDPC image (SSCC, fixed s_2)
Uncoded analog	uncoded analog image (JSCC-lite; DeepSC proxy)
DJSCC (learned)	compact SNR-conditioned deep JSCC image, our re-implementation
Fixed token	fixed detector tokens (GO-SG-style symbolic)
Evidence routing	(question, SNR) Wilson-LCB routing (ours)
Oracle	per-decision best of $\{s_1, s_2\}$ (selection upper bound)

larger than that of any feasible fixed policy—in particular it strictly Pareto-dominates the fixed-image policy: $A(r) \geq A(\pi_{s_2})$ and $E(r) < E(\pi_{s_2})$.

Proof sketch. (i) $A(\pi)$ separates over types and (E2) makes the rule a pointwise accuracy maximizer on every type. (ii) By the strict part of (E2) any accuracy-maximal policy agrees with r on T_{perc} ; where symbolic accuracies tie, (E1) makes the rule’s choice s_1 the cheaper one. (iii) π_{s_1} is infeasible at any $\tau > A(\pi_{s_1})$, the rule is feasible at every $\tau \leq A(r)$ by (i), and $E(r) < E(\pi_{s_2})$ follows from (E1) on the positive-mass set T_{sym} , with $A(r) \geq A(\pi_{s_2})$ by (i). $\square$

Remark 3 (Scope and empirical instantiation). *Like the lemmas above, Proposition 1 is conditional: (E1) and (E2) are measured properties of this system (Fig. 7 and Table III), not universal truths—if a future symbolic codec were more compute-hungry than the VLM, (E1) could fail. (E2) is instantiated with $T_{\text{sym}} = \{\text{counting, comparison, co_presence}\}$ and $T_{\text{perc}} = \{\text{presence}\}$, where both premises hold with margin at every SNR on the primary dataset; threshold sits outside the margin claim (its Δ CI crosses zero on both datasets and its sign flips across them, -0.015 vs. $+0.077$, Fig. 7(b)), so routing it to s_1 rests on (E1) alone—strictly cheaper, accuracy-neutral within the CI. Section VI-F reports the resulting frontier (rule $2.2\times$ cheaper and more accurate than fixed-image, Figs. 4 and 5). The proposition formalizes why routing, not coding, is the energy lever: it moves mass off the expensive evidence level wherever the cheap level is at least as accurate.*

VI. EXPERIMENTS

A. Setup and baselines

All numbers are test-set VQA accuracy on the deterministic 20% hold-out (Section III-H). The receiver VLM is Qwen2-VL-2B [20] unless stated; we compare the eight methods of Table I.

TABLE II

MAIN ACCURACY VS. SNR (RANGES SPAN $-5 \rightarrow 20$ dB), TEST SET, FIVE QUESTION TYPES. EVIDENCE ROUTING IS BEST NON-ORACLE EVERYWHERE. ERROR-FREE IMAGE REFERENCE = 0.611; “ADAPT. IMAGE” ABBREVIATES THE RATE-ADAPTIVE IMAGE BASELINE.

Channel	Routing	Token	Adapt. image	Fixed-rate ₋₅	Oracle ₂₀
AWGN	0.653 $\rightarrow$ 0.679	0.635	0.575 $\rightarrow$ 0.611	0.369	0.756
Rayleigh	0.641 $\rightarrow$ 0.679	0.633	0.561 $\rightarrow$ 0.612	0.409	0.755
Rician	0.651 $\rightarrow$ 0.681	0.635	0.584 $\rightarrow$ 0.611	0.369	0.758

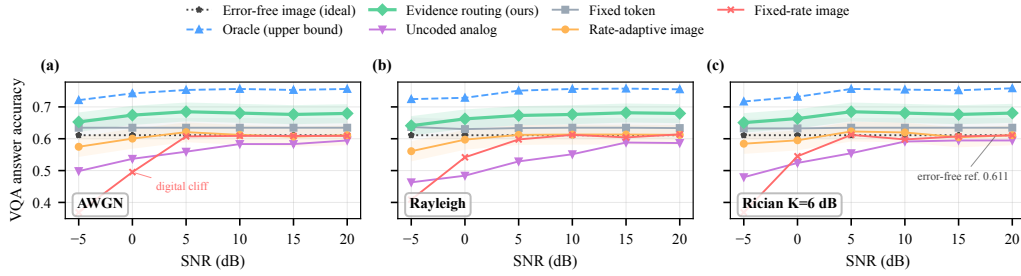


Fig. 2. Accuracy vs. SNR on the three channels (test set, five question types; Wilson 95% bands); evidence routing (ours) is best non-oracle at all SNRs; fixed-rate digital cliff vs. smooth analog/adaptive degradation.

The oracle is the per-decision maximum over the transmit-bearing levels $\{s_1, s_2\}$ (s_0 is a designed accuracy model, s_3 conditional, Section IV-D); the DJSCC baseline is trained and evaluated on the default Rician channel (Section VI-D).

B. Main result: accuracy vs. SNR

Table II reports accuracy across three channels and SNR extremes, over all five question types. Evidence routing is the best non-oracle method at every SNR on every channel; the analog baseline degrades smoothly (Rician $0.479 \rightarrow 0.595$) with no cliff. The headline: the router at 20 dB (0.679–0.681) *exceeds* the error-free image reference (0.611) by ≈ 7 points—routing beats full-image transmission even without channel errors, declining the image where it is the weaker evidence. One detail of Table II looks wrong but is real: at -5 dB the fixed-rate image is *higher* on Rayleigh (0.409) than on AWGN (0.369); Section VI-H traces it to the measured frame-error rates—fading upcrossings occasionally deliver frames a code AWGN holds below its waterfall deterministically. A second, milder inversion—rate-adaptive image 0.623 at 5 dB Rician vs. error-free 0.611—is not statistically significant (28 vs. 17 discordant, McNemar $p=0.14$) and concentrates on the yes-biased predicate types (co_presence and threshold, $+0.067$ each): the harsher 5 dB JPEG point (187 vs. 287 KB) filters low-salience clutter—the truncation mechanism of Section VI-N in JPEG form.

C. Sample size and significance

The primary evaluation comprises 936 unique test questions (presence 410, counting 214, and comparison, co_presence, threshold 104 each) from the deterministic 20% hold-out of the 548-image VisDrone pool, replayed across the six-point SNR grid: $936 \times 6 = 5616$ scored decisions per channel, 16,848 pooled; the 490-image DroneVehicle pool adds 710 test questions (4,260 Rician decisions). Since a question and its SNR replays are not independent, all intervals resample at the *image* level; with image-clustered bootstrap 95% CIs the transmit-bearing methods score, pooled over the three channels: rate-adaptive image 0.6040 [0.579, 0.631], fixed token 0.6341 [0.601, 0.667], evidence routing 0.6698 [0.640, 0.700], oracle 0.7459 [0.717, 0.776] (the per-channel spread across all three channels is at most 0.006, Table V). The router beats both fixed baselines decisively: paired McNemar versus the image $p < 10^{-103}$ (2690 discordant of the 16,848 pooled decisions), versus the token $p < 10^{-51}$ (1583).

D. Learned joint source–channel baseline (DJSCC)

Modern *learned* codecs are the fairer image-side alternative to the uncoded analog proxy; we add a deep-JSCC (DJSCC) baseline in the style of [15], [16]: a compact SNR-conditioned autoencoder (1.67M parameters, $1/32$ complex uses per source real, 9.2×10^4 uses per frame), trained on the *disjoint* VisDrone train split under the same post-ZF Rician block-fading abstraction and evaluated end-to-end with the same frozen VLM, prompts, and scoring; reconstruction is healthy for the size (mean PSNR 23.0 $\rightarrow$ 28.6 dB) with no cliff. Fig. 3 places it in the Rician grid, with three honest readings. (a) DJSCC behaves as learned JSCC should: it beats the analog proxy where noise dominates (0.521 vs. 0.479 at -5 dB), converges to it at high SNR, and is the cheapest image pipeline per channel use (2.5–41 $\times$ fewer uses). (b) It does *not* beat the rate-adaptive image at any SNR (0.564–0.600 vs. 0.584–0.623): a 1.67M codec at a fixed $1/32$ bandwidth ratio trades detail for robustness (no state-of-the-art claim, Section VII-B). (c) The routing conclusion is untouched: DJSCC counts no better than any image path (0.277), fixed tokens beat it at every SNR, and routing stays 7.6–9.3 points ahead—learned coding moves the image path’s *efficiency*, not the *which-evidence* answer.

E. Latency

The end-to-end latency decomposition makes the real-time argument for token routing under weak links: at -5 dB Rayleigh, rate-adaptive image transmission costs 3.72 s upload + 0.60 s VLM inference

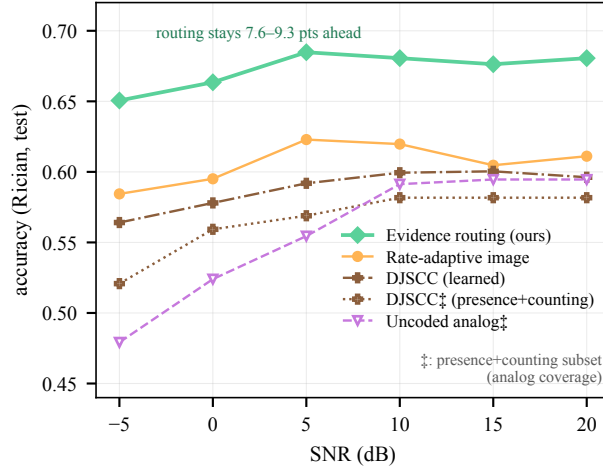


Fig. 3. Learned DJSCC baseline in the Rician test grid, accuracy vs. SNR (all five question types unless marked). Lines marked ‡ cover presence+counting only, the analog baseline’s coverage. DJSCC uses a constant 9.2×10^4 complex uses per frame.

TABLE III

MEASURED GPU POWER BY PHASE (RTX 4060 EDGE-SERVER PROXY; BETWEEN-WINDOW STD ≤ 1.1 W). INCREMENTAL = (PHASE – IDLE) POWER $\times$ LATENCY.

Phase (role)	$\bar{P}$ (W)	util	s/item	J/item incr.	total
idle (baseline)	52.3	0%	—	—	—
YOLOv8n (s_1 detector, cross-check)	58.7	18%	0.0104	0.066	0.61
Qwen2-VL-2B (s_2 answerer)	107.9	81%	0.581	32.3	62.7

= 4.32 s, while the token path completes in 0.089 s—a $49\times$ gap—and the fixed-rate image costs 5.87 s.

F. Measured power and the per-answer energy frontier

Table III reports the measured GPU power behind the compute term of Eq. (1): the driver’s internal ~ 8 Hz power sampler (`nvidia-smi`, 14.4 s trailing windows polled every 4 s) on the campaign RTX 4060 (115 W limit), declared as an edge-server proxy. Each phase discards a 20 s warm-up and averages window means over a steady-state loop of the *campaign workload itself*: for the VLM, 413 channel-degraded test images (5 dB Rician) with their real questions; for the detector, the s_1 pass over the test images; idle is re-measured after the loops (52.8 vs. 52.3 W). Energy per item is average power $\times$ measured per-item latency (the VLM’s 0.58 s/item matches Section VI-E’s 0.60 s); headline figures are *incremental over idle*, and the parameterized quantities of Section III-G complete Eq. (1) (P_{tx} headline 0.5 W; Jetson mid-band 0.43 J—the desktop cross-check, 0.066 J, falls below the band, conservative against the token path).

The frontier is compute-dominated. At every SNR and every P_{tx} in the band, the VLM forward pass costs more than the radio: the largest transmit energy of any image method is 2.7 J/answer at 0.5 W (5.3 J

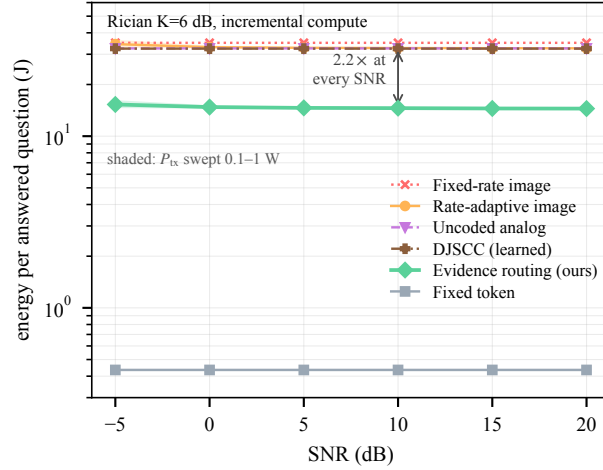


Fig. 4. Joint energy E per answered question vs. SNR, J (Rician; $P_{tx}=0.5$ W, incremental compute, Jetson mid-band detector; log energy axis); accuracies for the same methods and SNRs are in Figs. 2 and 3, paired with energy in Fig. 6(a). Shaded bands sweep P_{tx} over 0.1–1 W (3GPP aerial-UE class = 0.2 W); total-power billing raises image curves by ≈ 30.4 J without changing the ordering.

at 1 W), against 32.3 J of incremental VLM compute. Rate and power adaptation act on the small term; only evidence routing *declines the VLM forward pass*, the large one. The dominance assumes *unbatched, single-tenant* edge inference (Section III-G); server-style batching at $\sim 10\times$ would amortize the 32.3 J forward pass to ≈ 3 J—the order of the 2.7–5.3 J transmit term—turning dominance into parity, though never below the 0.435 J token path.

Figures 4 and 6 price every method under Eq. (1), and Fig. 5 decomposes each per-answer joule count into its radio, detector, and VLM components. Three readings. (a) Type-aware evidence routing strictly Pareto-dominates every fixed image pipeline, instantiating Proposition 1 with margin: $2.2\times$ less energy than the rate-adaptive image at every SNR (15.3 vs. 34.4 J at -5 dB; ratio 2.23–2.25 at each of the six SNRs, and point-identical on AWGN and Rayleigh) at +4.7 to +7 points *higher* accuracy; the learned DJSCC and uncoded analog baselines sit at the VLM floor (≈ 32.4 J) with lower accuracy still. (b) Fixed-token transmission is the cheap endpoint of the frontier: 0.435 J per answer—75–79 $\times$ below the rate-adaptive image, 33–35 $\times$ below routing—at 2–5 points below routing; nothing in the bank is both cheaper and more accurate. (c) In efficiency terms (Figure 6(b)), the fixed token delivers 1.46 correct answers per joule, routing 0.043–0.047, and every full-image method 0.011–0.019.

Sensitivity and conventions. Because the frontier is compute-dominated, no conclusion depends on the transmit-power parameterization: sweeping P_{tx} over 0.1–1 W at the worst link (shaded bands of Fig. 4)

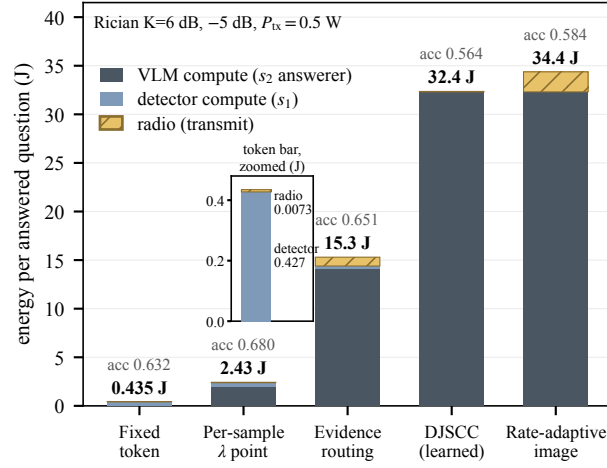


Fig. 5. Per-answer energy decomposition (Rician, -5 dB, $P_{tx}=0.5$ W, incremental billing): each evidence path split into radio, detector (s_1), and VLM (s_2) compute, with its measured accuracy above the bar. The per-sample λ point is the accuracy-saturation operating point of Section VI-G; unlike the single-SNR bars it is a whole-test-grid operating point (all six SNRs pooled, Rician), and its coordinates of record, 0.680 at 2.43 J, are what the text rounds to 2.4 J. Inset: the token bar rescaled—detector-dominated (0.43 of 0.435 J).

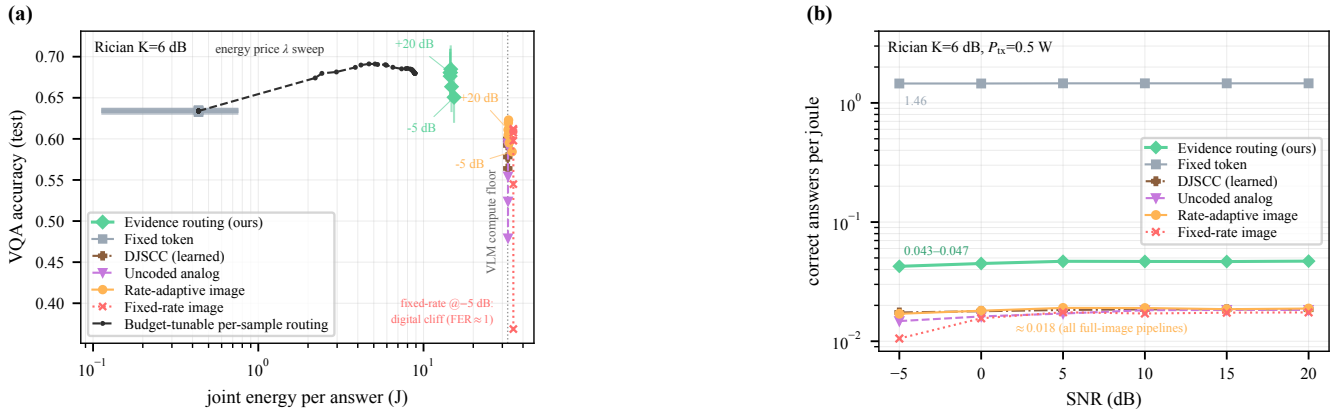


Fig. 6. Measured per-answer energy (Rician, $P_{tx}=0.5$ W, incremental compute, log energy axes). (a) Energy-accuracy frontier (fixed-token band = Jetson detector energy; dotted vertical guide = measured VLM compute floor; dashed curve: the budget-tunable per-sample router of Section VI-G, swept by the energy price λ). (b) Correct answers per joule vs. SNR.

moves the fixed token over 0.429–0.442 J, routing over 14.6–16.2 J, and the rate-adaptive image over 32.7–36.4 J, pinning the image/routing ratio at $2.2\times$ with every ordering intact; the same holds under total-power billing (routing 27.8–28.6 vs. image 62.9–64.8 J). Sanity check: the token path is detector-dominated (0.43 of 0.435 J at the band midpoint; 0.11–0.76 J across the declared Jetson band, 0.07 J desktop) and the image path VLM-dominated (32.4–35.0 J incremental, 62.8–65.4 total)—two orders of magnitude apart, matching the $265\times$ channel-use gap.

G. From energy-audited to energy-controllable routing

The frontier above prices *fixed* policies; with the sender-side predictors $P(\text{correct} \mid x, s)$ of Section IV-C (train split only) the operating point becomes tunable: each test sample is routed by $s^*(x) = \arg \max_s [P(\text{correct} \mid x, s) - \lambda E_s(\gamma)]$, with E_s the measured per-answer energy of Eq. (1); sweeping the energy price λ traces the dashed frontier of Figure 6. (a) The endpoints validate the construction: $\lambda \rightarrow \infty$ recovers the all-token point (0.435 J, 0.634), $\lambda=0$ the accuracy-first router of Table V (8.9 J, 0.680), and the all-image policy (33.0 J, 0.606) lies strictly below the curve. (b) Accuracy saturates early: 0.680 at 2.4 J—the accuracy-first level at $3.7\times$ less energy—peaking at 0.691 near 4.7 J (AWGN/Rayleigh peaks 0.694/0.690); with per-sample evidence admitted, even the type rule (0.67 at 14.6 J) is dominated. (c) The realized curve is non-monotone beyond ≈ 5 J—image-side predicted gains smaller than the price are mostly predictor noise, so a deployment would not operate below $\lambda \approx 3 \times 10^{-3} \text{ J}^{-1}$. As the dual of a per-answer energy budget, λ composes directly with slot-level resource allocation (Section VII-A).

H. Link-level footprints: residual frame loss and payload

The residual frame-loss probabilities of the two loss models (Section III-B), at the campaign operating points, explain two headline behaviors. *Why the token curves are flat:* the s_1 packet is tiny (measured mean 0.9 KB), so its information-outage probability stays below 0.7% on Rician and 5.2% on Rayleigh even at -5 dB—the token path almost never loses evidence. *Why the fixed-rate baseline falls off a cliff:* the measured post-decode FER of the rate-1/2 LDPC frames climbs from 1.4–36% at 5 dB to 98–100% at -5 dB, wiping out the image tiles; at -5 dB AWGN fails deterministically (FER 1.000) while Rayleigh occasionally delivers (FER 0.979)—the origin of the Rayleigh>AWGN inversion in Table II.

The image path behind Table V delivers the rate-adaptive byte budget $B\tau \cdot \text{SE}(\gamma)$ fitted by JPEG, growing from 14.0 KB at -5 dB to 215 KB at 20 dB (Rician; within 3–10% of budget at every SNR, deep-fade outage $\leq 1.3\%$), against a ~ 0.9 KB token packet—a $15\times$ – $236\times$ per-decision payload ratio that routing converts into the latency and energy gaps above. (The payload column of Table V is receiver-side stored-file bytes, a JPEG re-encode above the transmitted budget; the routing mix, not the codec, drives its halving.)

I. Evidence–question complementarity

Figure 7 reports $\Delta = \text{acc}(\text{token}) - \text{acc}(\text{image})$ by question type on both datasets (three channels pooled), with 95% CIs from an *image-clustered* bootstrap (5000 resamples; a question and its six SNR replays

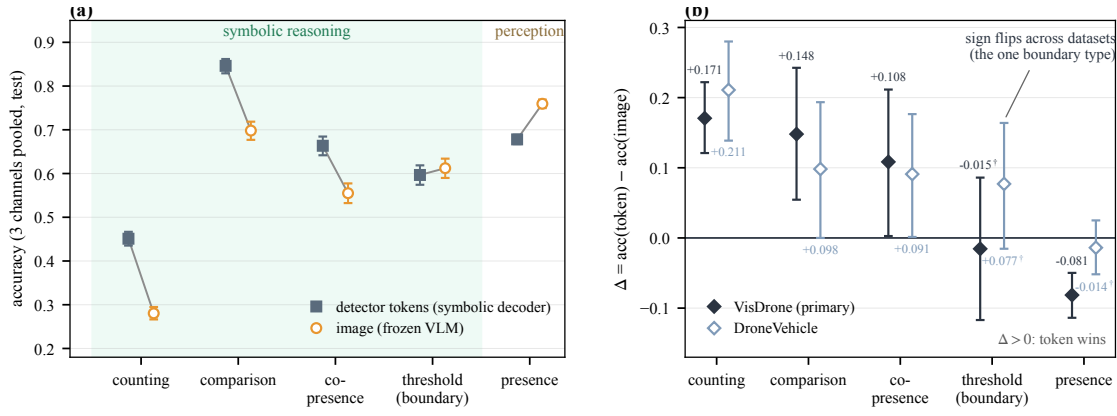


Fig. 7. The complementarity behind the router. (a) Per-type accuracy of the two transmit-bearing levels (VisDrone, three channels pooled, test split, $n=16,848$; Wilson 95% intervals); the symbolic types (counting, comparison, *co_presence*) are token-favored, presence is perception-favored, threshold is the boundary type. (b) The gap Δ per type on both datasets, with image-clustered bootstrap 95% CIs (5000 resamples) and the numeric Δ of record at each point; † marks CIs that cross zero (threshold on both datasets, presence on DroneVehicle). The sign pattern agrees on 4/5 types across datasets; threshold is the one flip.

move together). Counting and comparison exclude zero on both datasets, and presence is solidly image-favored on VisDrone. Two types are genuinely marginal: *co_presence* is sign-consistent (token-favored) but its VisDrone interval barely clears zero; *threshold* is a *boundary* type whose interval crosses zero on both datasets and whose sign flips across them (-0.015 vs. $+0.077$)—the one type where routing is not clear-cut (Section VI-J).

J. Cross-VLM robustness

Fig. 8 repeats the per-type split for three receiver models spanning two families: Qwen2-VL-2B, Qwen2.5-VL-3B [31], and the architecturally distinct SmolVLM-Instruct [32] (Rician); the token bar is the VLM-agnostic symbolic decoder, identical by construction. The routing direction agrees on 4/5 types; the only divergence is threshold, the boundary type with the smallest $|\Delta|$. The stronger 3B model reads degraded images markedly better (counting 0.375 vs. 0.280), yet the routing is unchanged: the complementarity is structural, not receiver-specific.

K. Who should decode the tokens?

Could the token advantage be an artifact of answering symbolic questions with a calibrated symbolic decoder rather than the VLM? We test this directly: the receiver VLM is handed the *same channel-degraded token text* and answers with no image input (Rician, $\text{SNR} \in \{-5, 5, 20\}$ dB; 2808 cells). Table IV reports the outcome: the VLM is the *worst* consumer of the tokens on every question type—pooled 0.485

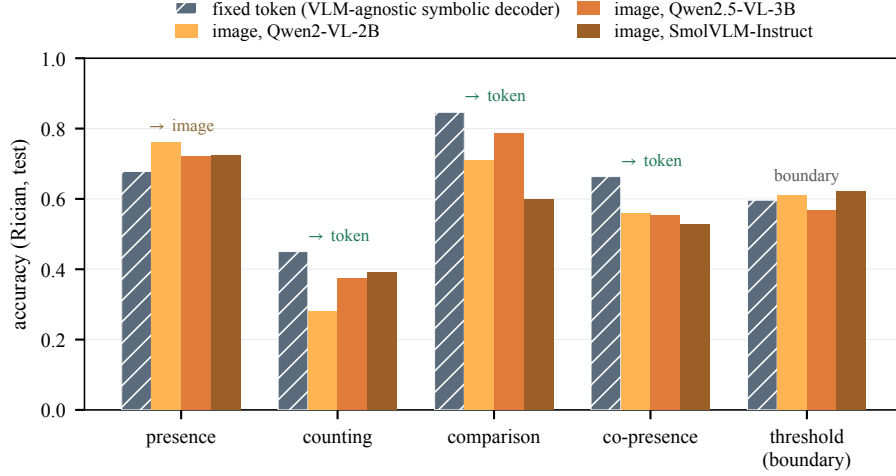


Fig. 8. Cross-VLM per-type accuracy (Rician): one VLM-agnostic token bar per type (hatched), three image bars (one per receiver VLM). The routing verdict annotated over each group agrees on 4/5 types; threshold is the boundary type.

vs. 0.634 for the symbolic decoder (-14.9 points; paired McNemar $p < 10^{-27}$)—with the largest gap on comparison (0.532 vs. 0.846), whose count arithmetic the VLM fails to reproduce from text. The token *content* does help the VLM where the degraded image is poor (counting: 0.364 vs. 0.273), but calibration plus deterministic decoding converts content into accuracy (0.450). The edge-side symbolic decoder is therefore a load-bearing design choice—and what makes the token bars of Fig. 8 receiver-independent.

One objection remains: questions and parser share the same templates, so the decoder’s edge could be template alignment rather than answering ability. A free-phrasing probe tests this: 440 held-out questions each receive two free-form paraphrases (880 items; Rician, 5 dB), replayed through the same three consumers. The *answering* advantage is phrasing-robust: the pooled decoder-over-VLM-reads-tokens gap is $+0.165$ under paraphrase (0.632 vs. 0.467; McNemar $p < 10^{-3}$) vs. $+0.159$ on original wording, sign preserved on all five types (largest: comparison, $+0.30$). The question-*parsing* front-end is not: the strict template parser fails on 100% of paraphrases; a lightweight keyword intent classifier restores parsing (0% failure, 100% intent match), after which per-type decoder accuracy is identical to original wording (zero discordant pairs per type). Both VLM channels are phrasing-robust (reads-tokens $0.473 \rightarrow 0.467$, $p=0.60$; reads-image $0.580 \rightarrow 0.551$, $p=0.022$, mild). The advantage is thus answering, not template alignment; free-form deployment needs the intent front-end (Section VII-B).

TABLE IV
TOKEN-CONSUMER ABLATION (RICIAN, SNR $\in \{-5, 5, 20\}$ dB, TEST SET): THE SAME TRANSMITTED TOKENS, CONSUMED THREE WAYS. BOLD MARKS THE BEST CONSUMER PER ROW.

Type	n	VLM reads tokens	symbolic decoder	VLM reads image
presence	1230	0.554	0.678	0.761
counting	642	0.364	0.450	0.273
comparison	312	0.532	0.846	0.712
co_presence	312	0.449	0.663	0.574
threshold	312	0.446	0.596	0.609
pooled	2808	0.485	0.634	0.606

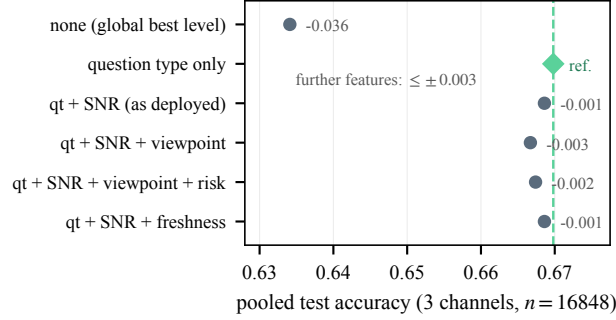


Fig. 9. Routing-dimension ablation (LCB selector, pooled test accuracy over three channels, $n=16,848$; the per-point labels give Δ vs. the question-type-only reference). Question type carries the decision (0.670 vs. 0.634 unconditioned); further features add $\leq \pm 0.003$.

L. Which routing dimensions matter

Fig. 9 ablates the conditioning dimensions of the LCB selector under the identical train/test split. Question type alone recovers essentially all achievable routing accuracy (0.670 pooled); adding SNR, viewpoint, freshness, or risk changes pooled test accuracy by at most ± 0.003 (SNR and viewpoint slightly lower it)—the empirical basis for the zero-parameter rule: question type is the only load-bearing routing dimension (the rest belong to the scheduling layer, Section VII-A).

A consistency check: since the deployed rule ignores SNR, a scheduling-layer CSI error cannot hurt the routing; a full policy-assumed $\times$ true-SNR mismatch matrix (the physical link still adapts to the true SNR) confirms it—all three channels nearly flat, the worst mismatched cell (0.641–0.653) equaling the lowest matched diagonal.

M. Per-sample predictor

Table V evaluates the sender-side logistic predictor (Section IV-C) as offline re-selection. It beats the LUT on all three channels by +0.7 to +1.1 pt while nearly halving mean payload.

TABLE V

PER-SAMPLE PREDICTOR VS. LUT ROUTING (ACCURACY; “IMAGE” IS THE RATE-ADAPTIVE IMAGE; MEAN PER-DECISION PAYLOAD, MEASURED AS RECEIVER-SIDE DELIVERED-IMAGE BYTES—THE TRANSMITTED PER-SNR BUDGETS ARE IN SECTION VI-H). THE CLUSTERED CIs AND McNEMAR TESTS OF SECTION VI-C POOL THE THREE ROWS OF THIS GRID ($n=16,848$), WITH ROUTING EVALUATED AS THE TYPE RULE OF SECTION IV-B (0.6698 POOLED). THE LUT COLUMN IS CALIBRATED *per channel* (EACH CHANNEL’S OWN CALIBRATION SPLIT), WHEREAS THE POOLED SELECTOR FIGURE OF 0.6686 QUOTED IN THE TEXT IS A SINGLE POLICY CALIBRATED ON THE POOLED THREE-CHANNEL TABLE; THE TWO CALIBRATION REGIMES THEREFORE NEED NOT AVERAGE TO THE SAME VALUE.

Channel	Token	Image	LUT	Per-sample	Oracle	Payload
AWGN	0.6346	0.6043	0.6747	0.6832	0.7472	58/111 KB
Rayleigh	0.6339	0.6015	0.6690	0.6804	0.7455	56/107 KB
Rician	0.6339	0.6063	0.6727	0.6798	0.7450	60/110 KB

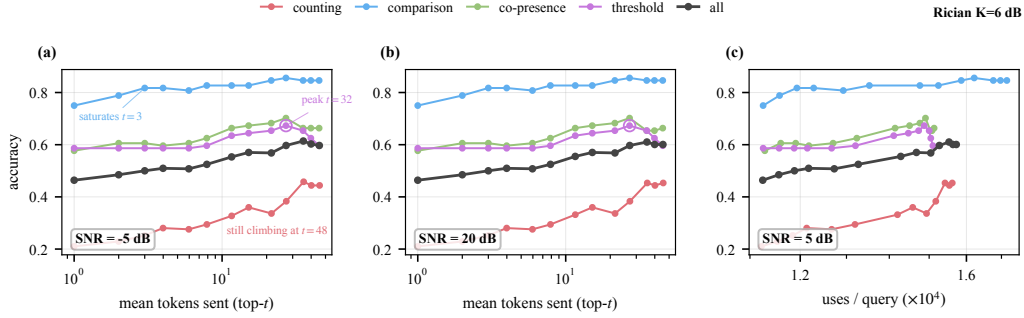


Fig. 10. Top- t token-budget sweep (symbolic types): comparison saturates at $t=3$; counting needs $t=48$; threshold peaks at $t=32$. Panels (a) and (b) plot accuracy against the mean number of tokens sent at the two SNR extremes (-5 and 20 dB); panel (c) replots the mid-SNR (5 dB) sweep against complex channel uses per query ($\times 10^4$), the airtime the budget knob actually buys.

N. Variable token budget

Figure 10 scans the top- t budget for symbolic types (each t re-calibrated; $t=\text{full}$ matches the main experiment bit-for-bit). Two findings. (A) *Budget sensitivity diverges by an order of magnitude*: comparison is near-saturated at $t=3$ (0.817 vs. full 0.846) while counting needs $t=48$ to reach 0.453—a second axis of complementarity. (B) *Less can be more*: threshold reaches 0.673 at $t=32$ vs. 0.596 at full (+7.7 pt) and co_presence gains +3.9 pt on all three channels: confidence-ordered truncation filters false positives on yes-biased questions. The header dominates the s_1 payload (696 B at $t=1$ vs. ≈ 1.2 KB full), so truncation saves only $\sim 30\%$ bandwidth—the knob’s value is accuracy filtering.

Reproducibility and run-to-run variability. The `crc32` image-hash split (Section III-H) makes the hold-out bit-reproducible; the scripts, calibration/LUT artifacts, and per-decision logs are version-controlled. An independent recomputation on different hardware agrees point-by-point on the shared two-type subset (AWGN routing $0.628 \rightarrow 0.668$ vs. $0.627 \rightarrow 0.668$); two end-to-end s_1 reruns agree on 99.15% of test verdicts (aggregate 0.6709 both); and ten channel-seed redraws of the two most-cited cells give 0.581 ± 0.003 (rate-adaptive image) and 0.372 ± 0.003 (fixed-rate) at -5 dB Rician—seed noise an order

of magnitude below every interpreted gap.

VII. DISCUSSION

A. A safety bridge to UAV traffic management

The latency gap of Section VI-E also has an airspace-safety counterpart: in U-space separation management, where the tactical separation minimum grows with communication delay (BUBBLES [33]; Weinert *et al.* [34]), an M/G/1 analysis of the mission uplink (companion scheduling study) finds that token routing leaves the certified 2σ separation minimum at baseline (370.6 vs. 370.5 m at -5 dB Rayleigh), whereas continuous image upload inflates it to 394.3 m at peak load and fixed-rate imagery to ≈ 404 m—the same evidence frugality that saves energy (Section VI-F) preserves the certified separation baseline. More broadly, the router exposes the interface a scheduling layer needs—a discrete evidence-level action with a measured accuracy lower bound—and SNR/freshness/risk informing resources but *not* the level decision (Section IV-B) is why the layers separate cleanly.

B. Limitations

We list the honest boundaries of the study. (i) *Single-channel second dataset*: the DroneVehicle replication is Rician-only, so the cross-dataset claim is strongest there. (ii) *Compact learned baseline, not state-of-the-art*: the learned image baseline (DJSCC) is a compact 1.7M-parameter re-implementation—a controlled reference, not state-of-the-art; a purpose-built codec such as VideoQA-SC [11] could drop in as an additional learned compact-evidence level—an alternative to the coded image s_2 —within our routing framework. (iii) *Scale*: the 548+490-image pool (16,848+4,260 scored test decisions), while the largest such VLM-over-wireless VQA evaluation we know of, is an order of magnitude below classification-scale semantic-communication studies; a broader receiver survey is future work. (iv) *Analytical idealization*: the lemmas of Section V are *exact* only in idealized limits, directional outside them. (v) *Template-generated questions*: questions are programmatically generated from detection ground truth; a free-phrasing probe (Section VI-K) shows the answering advantage survives paraphrase (+0.165 vs. +0.159), but the strict parser is template-bound (a keyword intent classifier restores parsing); open-vocabulary human phrasing remains untested. (vi) *Offline gains need not transfer to closed loop*: the per-sample predictor (Section IV-C) is an *offline re-selection* winner (the energy-price frontier of Section VI-G inherits this), but in preliminary closed-loop scheduling (companion study) the LUT selector wins once resource and

safety feedback enter—offline optimum $\neq$ closed-loop optimum. (vii) *Energy proxies*: VLM power is measured on a desktop-class GPU declared as an edge-server proxy, the onboard detector is a parameterized Jetson-class band, and P_{tx} is a band anchored to the 3GPP aerial-UE class (Section III-G); the $>10\times$ compute-dominance margin absorbs these proxies.

VIII. CONCLUSION

We argued that an energy-frugal UAV VQA transmitter should ask, before it transmits, which *level* of evidence the question needs—that decision selects not only the payload but whether the most energy-hungry step, the VLM forward pass, runs at all. We characterized an evidence–question complementarity—symbolic questions favor tokens, perception questions the image—and showed that a zero-parameter rule matches a data-calibrated selector (bit-identical on the primary dataset, statistically indistinguishable on a second), with two conditional lemmas explaining why; the token advantage requires the calibrated symbolic decoder (tokens as VLM text cost ≈ 15 points). On the largest VLM-over-wireless VQA evaluation we are aware of (548+490 images scored as 16,848 held-out test decisions plus a 4260-decision replication; five question types, eight methods, unified channel-use *and* measured joint-energy accounting), evidence routing beat full-image transmission even over an error-free channel, was CSI-insensitive by construction, and benefited from a non-monotone token-truncation effect that filters false positives. The measured per-answer frontier is compute-dominated, and type-aware routing Pareto-dominates every fixed image pipeline— $2.2\times$ less energy than rate-adaptive transmission at higher accuracy, with fixed tokens $75\text{--}79\times$ cheaper than the image ($33\text{--}35\times$ cheaper than routing)—so in this regime energy efficiency comes from evidence frugality, not from better source or channel coding alone. An energy price swept over the per-sample predictor turns the audited frontier into a *controllable* one: a single scalar spans all-token to accuracy-first operation and, being a per-answer energy-budget dual, composes directly with slot-level resource allocators. Future work: the two proofs, all-channel and non-Qwen extension of the second dataset, a learned symbolic codec, embedded detector-power measurement, and closing the loop to dynamic separation.

REFERENCES

- [1] D. Gündüz, Z. Qin, I. E. Aguerri *et al.*, “Beyond transmitting bits: Context, semantics, and task-oriented communications,” *IEEE Journal on Selected Areas in Communications*, vol. 41, no. 1, pp. 5–41, 2023.
- [2] H. Xie, Z. Qin, G. Y. Li, and B.-H. Juang, “Deep learning enabled semantic communication systems,” *IEEE Transactions on Signal Processing*, vol. 69, pp. 2663–2675, 2021.

- [3] M. Kalfa, M. Gok, A. Atalik, B. Tegin, T. M. Duman, and O. Arikan, "Towards goal-oriented semantic signal processing: Applications and future challenges," *Digital Signal Processing*, vol. 119, p. 103134, 2021.
- [4] G. Zheng, Q. Ni, K. Navaie, H. Pervaiz, A. Kaushik, and C. C. Zarakovitis, "Energy-efficient semantic communication for aerial-aided edge networks," *IEEE Transactions on Green Communications and Networking*, vol. 8, no. 4, pp. 1742–1751, 2024.
- [5] J. Dai, J. Li, Z. Zhao, Z. Yang, Z. Zhang, and M. Shikh-Bahaei, "Energy efficient multi-modal probabilistic semantic communication (PSCoM)," *IEEE Transactions on Green Communications and Networking*, vol. 9, no. 4, pp. 1951–1963, 2025.
- [6] R. Xu, Z. Yang, Y. Mao *et al.*, "Rate-splitting multiple access enabled green probabilistic semantic communication over wireless networks," *IEEE Transactions on Green Communications and Networking*, vol. 9, no. 4, pp. 1472–1486, 2025.
- [7] H. Xie, Z. Qin, X. Tao, and K. B. Letaief, "Task-oriented multi-user semantic communications," *IEEE Journal on Selected Areas in Communications*, vol. 40, no. 9, pp. 2584–2597, 2022, arXiv:2112.10255.
- [8] S. Liu, N. Li, Y. Deng, and T. Q. S. Quek, "Goal-oriented semantic communication for wireless visual question answering," 2024, arXiv:2411.02452.
- [9] B. Du, H. Du, D. Niyato, and R. Li, "Task-oriented semantic communication in large multimodal models-based vehicle networks," *IEEE Transactions on Mobile Computing*, vol. 24, no. 10, pp. 9822–9836, 2025, arXiv:2505.02413.
- [10] H. Xie, Z. Qin, and G. Y. Li, "Task-oriented multi-user semantic communications for VQA," *IEEE Wireless Communications Letters*, vol. 11, no. 3, pp. 553–557, 2022, arXiv:2108.07357.
- [11] J. Guo, W. Chen, Y. Sun, J. Xu, and B. Ai, "VideoQA-SC: Adaptive semantic communication for video question answering," *IEEE Journal on Selected Areas in Communications*, vol. 43, no. 7, pp. 2462–2477, 2025, arXiv:2406.18538.
- [12] J. Park and S. W. Yoon, "Transmit what you need: Task-adaptive semantic communications for visual information," *IEEE Journal on Selected Areas in Communications*, vol. 43, no. 12, pp. 4182–4197, 2025.
- [13] C. S.-H. Hsu, W. Y.-C. Cheng, and C. Papagianni, "Progressive semantic communication for efficient edge-cloud vision-language models," 2026, arXiv:2604.26508.
- [14] W. Zhang, H. Zhang, H. Ma, H. Shao, N. Wang, and V. C. M. Leung, "Predictive and adaptive deep coding for wireless image transmission in semantic communication," *IEEE Transactions on Wireless Communications*, vol. 22, no. 8, pp. 5486–5501, 2023.
- [15] E. Bourtsoulatz, D. B. Kurka, and D. Gündüz, "Deep joint source-channel coding for wireless image transmission," *IEEE Transactions on Cognitive Communications and Networking*, vol. 5, no. 3, pp. 567–579, 2019.
- [16] J. Xu, B. Ai, W. Chen, A. Yang, P. Sun, and M. Rodrigues, "Wireless image transmission using deep source channel coding with attention modules," *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 32, no. 4, pp. 2315–2328, 2022.
- [17] P. Joshi and M. Gurusamy, "Context-aware information transfer via digital semantic communication in uav-based networks," 2026, arXiv:2601.01430.
- [18] D. Won, Q. T. Do, T. T. Win, D. Lee, J. Oh, and S. Cho, "Multidomain adaptive semantic communications," *IEEE Journal on Selected Areas in Communications*, vol. 43, no. 7, pp. 2506–2517, 2025.
- [19] G. Jocher, A. Chaurasia, and J. Qiu, "Ultralytics YOLOv8," <https://github.com/ultralytics/ultralytics>, 2023.
- [20] P. Wang, S. Bai, S. Tan *et al.*, "Qwen2-vl: Enhancing vision-language model's perception of the world at any resolution," 2024, arXiv:2409.12191.
- [21] R. G. Gallager, "Low-density parity-check codes," *IRE Transactions on Information Theory*, vol. 8, no. 1, pp. 21–28, 1962.
- [22] Y. Polyanskiy, H. V. Poor, and S. Verdú, "Channel coding rate in the finite blocklength regime," *IEEE Transactions on Information Theory*, vol. 56, no. 5, pp. 2307–2359, 2010.
- [23] D. Tse and P. Viswanath, *Fundamentals of Wireless Communication*. Cambridge University Press, 2005.
- [24] W. Khawaja, I. Guvenc, D. W. Matolak, U.-C. Fiebig, and N. Schneckenburger, "A survey of air-to-ground propagation channel modeling for unmanned aerial vehicles," *IEEE Communications Surveys & Tutorials*, vol. 21, no. 3, pp. 2361–2391, 2019.
- [25] A. Al-Hourani, S. Kandeepan, and S. Lardner, "Optimal LAP altitude for maximum coverage," *IEEE Wireless Communications Letters*, vol. 3, no. 6, pp. 569–572, 2014.
- [26] 3GPP, "Study on enhanced LTE support for aerial vehicles," 3rd Generation Partnership Project, Tech. Rep. TR 36.777 V15.0.0, 2017, aerial UE transmit power class 23 dBm.
- [27] NVIDIA Corporation, "Jetson Orin Nano series modules data sheet," <https://www.nvidia.com/en-us/autonomous-machines/embedded-systems/jetson-orin/>, 2024, original power modes: configurable module power 7–15 W.
- [28] P. Zhu, L. Wen, D. Du *et al.*, "Detection and tracking meet drones challenge," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 44, no. 11, pp. 7380–7399, 2022.
- [29] Y. Sun, B. Cao, P. Zhu, and Q. Hu, "Drone-based RGB-infrared cross-modality vehicle detection via uncertainty-aware learning," *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 32, no. 10, pp. 6700–6713, 2022.
- [30] E. B. Wilson, "Probable inference, the law of succession, and statistical inference," *Journal of the American Statistical Association*, vol. 22, no. 158, pp. 209–212, 1927.
- [31] S. Bai, K. Chen, X. Liu, J. Wang, W. Ge *et al.*, "Qwen2.5-vl technical report," 2025, arXiv:2502.13923.
- [32] A. Marafioti, O. Zohar, M. Farré, M. Noyan, E. Bakouch *et al.*, "SmolVLM: Redefining small and efficient multimodal models," 2025, arXiv:2504.05299.
- [33] BUBBLES Consortium, "D2.1 concept formulation for U-space separation management service," SESAR Joint Undertaking, Grant No. 893206, Tech. Rep. Ed. 04.00.00, 2022.
- [34] A. Weinert, L. Alvarez, M. Owen, and B. Zintak, "Near midair collision analog for drones based on unmitigated collision risk," *Journal of Air Transportation*, vol. 30, no. 2, pp. 37–48, 2022.